

Report 1: The State of Rural Agriculture and Household Income

Charitarth Foundation Deep Research Series — Pillar I: Agriculture to Rural Economy Transition

Executive Summary

Indian agriculture presents a paradox that sits at the centre of the rural employment debate. Nearly half the country's workforce still depends on farming for its livelihood, yet the sector produces less than a fifth of national output. That gap, more than any single crop failure or price shock, is the starting point for understanding why rural households remain economically fragile even as the aggregate economy grows.

This report establishes the baseline facts: how many people work in agriculture relative to what it produces, how farm households actually assemble their income, how fragmented Indian landholding has become, what limits productivity, and how water availability shapes all of it. Each section flags the strength of the underlying evidence, and closes by testing the report's central question directly: is rural India's problem a shortage of jobs, or a shortage of productivity within the jobs that already exist?

1. The Workforce-Output Gap

Agriculture and allied activities (crop farming, livestock, forestry and fishing) contributed about 18 percent of India's Gross Value Added at current prices in 2024-25, with nominal GVA from the sector crossing roughly ₹54 lakh crore. At constant (2011-12) prices the sector's share was closer to 14 percent. This share has fallen consistently for decades, down from over half of GDP in the early 1950s. **[Evidence: A]**

Employment has not fallen in step. Per the Periodic Labour Force Survey, agriculture's share of total employment stood at 44.1 percent in 2017-18 and had actually risen to 46.1 percent by 2023-24, a reversal of the long expected structural shift out of farming. Manufacturing's employment share fell over the same period, from 12.1 to 11.4 percent, and services fell from 31.1 to 29.7 percent. Rural labour force participation itself jumped sharply, from about 50.7 percent in 2017-18 to 63.7 percent in 2023-24, with the increase driven overwhelmingly by women entering the workforce (rural female LFPR rose from roughly 24 percent to 42 percent over the same window). Much of this reflects a "return to the farm" during and after the pandemic rather than agriculture generating genuinely new, better-paid work. **[Evidence: A]**

The arithmetic is stark: a sector employing roughly 46 percent of workers is generating under 18 percent of output. By definition, average output per worker in agriculture is a fraction of average output per worker elsewhere in the economy. This is the single most important structural fact governing rural income in India, and it reframes the standard question. Rural India's difficulty is not simply that there are "not enough jobs." A large share of the workforce already has a job, in the sense of being occupied in agricultural

activity. The deeper problem is that too much labour is doing too little economically productive work per person, often on holdings too small to fully occupy or fully reward the people working them. This is consistent with long-documented patterns of disguised unemployment and underemployment in Indian agriculture, where family members are recorded as "working" on the farm even though their marginal contribution to output is very low. **[Evidence: B — the qualitative pattern of disguised unemployment is well established in the literature; a precise, current national estimate of how much of the 46 percent is genuinely productive versus disguised is not something this report can quantify with confidence.]**

The rise in women's recorded agricultural participation deserves a separate flag: it may partly reflect better survey capture of unpaid family labour that always existed, rather than a real increase in women's paid economic activity. This distinction matters enormously for Pillar II of this research programme and is treated in more depth there.

2. How Agricultural Households Actually Earn Money

The National Statistical Office's Situation Assessment Survey (SAS) of Agricultural Households, most recently conducted in 2019 for the agricultural year July 2018 to June 2019, is the best available source on the composition of farm income. Average monthly income per agricultural household was estimated at ₹10,218, up from ₹6,426 in the 2013 survey, an increase of about 59 percent in nominal terms. **[Evidence: A]**

The composition of that income is the more important finding. Crop cultivation, once the dominant source of farm household income, contributed only about 37 percent of the total in 2019, down from around 48 percent in 2013. Wages and salaries have overtaken cultivation as the largest single component, at roughly 40 percent. The remainder comes from livestock, leasing out land, and non-farm business activity. **[Evidence: A]**

This is a significant and under-appreciated shift. The average "agricultural household" in India today is no longer primarily a cultivator household; it is a multi-income household in which farming is one activity among several, and often not the largest income earner. Households increasingly combine casual agricultural wage labour, non-farm wage labour, small livestock enterprises and occasional non-farm business income to construct a viable livelihood, because cultivation alone on a typical small holding does not generate enough income or enough employment days in a year. Livestock and allied activities have grown faster than crop farming in value terms and now play a stabilising, income-smoothing role across the agricultural calendar. **[Evidence: A/B: the income shares are well documented; the income-smoothing role of livestock across seasons is a widely supported inference in the agricultural economics literature rather than a single directly measured statistic.]**

Two implications follow directly for Charitarth's design choices. First, any intervention that treats "farmer" and "cultivator" as synonymous risks missing the reality that most agricultural households already look outward for income, toward wage labour and allied activity, and design should meet households where their actual income portfolio already sits. Second, income diversification is not a symptom of agricultural failure to be

reversed; for a large share of small and marginal households it is the rational and necessary response to a landholding base that cannot support a family on cultivation income alone.

3. Landholding Structure: The Fragmentation Constraint

Land fragmentation is the physical constraint underneath both of the patterns above. As per the 2015-16 Agriculture Census (the most recent full census with published results), the average size of an operational holding in India had fallen to 1.08 hectares, down from 1.15 hectares in 2010-11 and from 2.28 hectares in 1970-71. Small and marginal holdings (below 2 hectares) made up 86 percent of all operational holdings by number, yet accounted for only about 47 percent of total operated area. Marginal holdings alone (below 1 hectare) constituted roughly 68.5 percent of all holdings. At the other extreme, holdings of 10 hectares and above were just 0.6 percent of the total but still commanded about 9 percent of operated area. **[Evidence: A]**

The distribution is heavily concentrated regionally: roughly half of India's small and marginal farmers live in just five states (Uttar Pradesh, Bihar, Maharashtra, Madhya Pradesh and Andhra Pradesh), while average holding size varies enormously across states, from under 0.2 hectares in Kerala to nearly 5 hectares in Nagaland. This means any national threshold for "viable farm size" is close to meaningless; viability depends on soil quality, irrigation, cropping pattern and local wage rates as much as on hectares. **[Evidence: A]**

Two consequences of fragmentation matter most for this research programme. Smaller holdings make mechanisation, on-farm storage, drip irrigation and direct market access individually uneconomic, since the fixed costs of most of this infrastructure do not scale down proportionally with plot size; this is the core economic argument for aggregation models (FPOs, custom hiring centres, shared processing) rather than household-by-household investment. Second, the continuing decline in average holding size, driven by inheritance-based subdivision, means that for a growing share of households, cultivation income alone is structurally incapable of supporting the family regardless of how much productivity improves on that plot. This is the landholding-side explanation for why wage income and non-farm income have overtaken crop income in the SAS data above.

4. Productivity Constraints

India's cereal yields, particularly rice and wheat, generally lag well behind top global performers and behind the best-performing Indian districts, though the size of that gap varies enormously by crop, irrigation source and region; closing it fully is neither realistic nor, in water-stressed regions, environmentally desirable. Groundwater-irrigated plots consistently out-yield surface-water-irrigated and rainfed plots for the same crop, because groundwater allows farmers to control timing and volume of application rather than depending on canal schedules or rainfall. **[Evidence: B — the direction and general magnitude of these gaps is well supported; a single precise "India versus**

world" yield-gap number is not something this report treats as settled, since estimates vary by methodology and crop.]

The constraints behind these gaps are well known individually, even if their relative weight differs by region: input quality and access (seed, fertiliser, soil health), extension service reach (technical advice reached under half of agricultural households even before a further decline reported between 2018 and 2019), credit access, mechanisation, and market price incentives that do not always reward quality or reduce post-harvest loss. **[Evidence: A/B]**

Rather than treating productivity as one undifferentiated problem, the practical question for Charitarth is which specific constraint binds in a given district and crop: is it seed and soil health, is it access to timely irrigation, is it extension and knowledge, or is it that the market simply does not pay enough for a better-produced crop to justify the additional investment. Districts differ, and any district diagnostic (see Part LX of the master research brief) should test each of these separately rather than assuming a single universal fix.

5. Irrigation and Water: The Crop-Choice Trap

India's net irrigated area is roughly 63 million hectares against roughly 140 million hectares of net sown area, meaning a large share of Indian farming remains rain-dependent. Agriculture accounts for the large majority of the country's total water use, commonly cited at close to four-fifths of freshwater consumption, concentrated heavily in water-intensive rice and wheat cultivation. **[Evidence: A/B]**

The rice-wheat cropping system that anchored India's food security since the Green Revolution is now the primary driver of groundwater depletion in the northern plains, with published estimates putting the loss at several hundred cubic kilometres of groundwater over the past two decades in the worst-affected states. Existing price supports (MSP) and procurement systems continue to reward rice and wheat cultivation even in regions where the water cost of doing so is now unsustainable, creating a structural incentive that pulls against water-efficient crop diversification. **[Evidence: A/B]**

This is directly relevant to Charitarth's crop-diversification thinking (Part VII of the master brief): pushing farmers toward higher-value horticulture or pulses on genuinely water-stressed land is not just an income-diversification opportunity, it is frequently the more sustainable path, but it will only succeed where market demand, storage and processing exist to absorb a more perishable, less policy-protected crop. Diversification recommended without that market-side infrastructure risks leaving farmers more exposed to price risk than they were under a rice-wheat system with assured procurement, even if the water economics clearly favour a change.

6. Synthesis: A Productivity Problem, Not Simply an Employment Problem

Returning to the report's central question: the evidence assembled here supports the view that rural India's difficulty is substantially a productivity and income-adequacy problem occurring within existing work, rather than purely a shortage of things for people to do. Nearly half the workforce is already occupied in agriculture; the trouble is that this occupation, on holdings averaging close to a hectare, cannot generate enough income to support a household, which is precisely why agricultural households now earn more from wages than from their own cultivation.

This has a direct implication for where Charitharth should look for leverage. Interventions aimed only at "more farming, done better" will help but will not resolve the underlying arithmetic for the roughly 86 percent of holders working under 2 hectares. The more consequential opportunities lie in the stages after harvest, aggregation, storage, processing, and in the non-farm rural enterprise and wage economy that already provides 40 percent-plus of agricultural household income today. This finding sets up the analysis in Report 2 of this series, which follows the agricultural value chain from farm gate to consumer to identify exactly where that non-farm value and employment could be created closer to the producing district.

Evidence Quality Notes and Open Questions

- Landholding data (Section 3) is drawn from the 2015-16 Agriculture Census. The 11th Agriculture Census (reference year 2021-22) was launched in 2022; where its results become available, the fragmentation figures in this report should be refreshed, as the direction of change (further fragmentation) is very likely to continue but the precise current figures may differ.
- Farm household income figures (Section 2) are from the 2019 Situation Assessment Survey, now several years old. NITI Aayog's earlier recommendation to run this survey every five years, rather than roughly once a decade, has not yet resulted in a newer round at the time of this report; any newer SAS release should be treated as the priority update.
- Precise India-versus-international yield gap figures (Section 4) vary by source and methodology; this report has deliberately avoided citing a single number as definitive.
- District-level and crop-level detail behind every figure in this report is a research priority for Part LX of the master brief (District Value-Chain Diagnostic) and should be built from state agriculture department and district statistical handbook data rather than national averages, since state and district variation is large enough that national figures can mask very different local realities.

Key Sources

- National Statistical Office (MoSPI), Periodic Labour Force Survey, Annual Reports 2017-18 through 2023-24
- Economic Survey of India, 2023-24 and 2024-25 editions (Ministry of Finance)

- NSS Report No. 587, Situation Assessment of Agricultural Households and Land and Livestock Holdings of Households in Rural India, 2019 (77th Round), National Statistical Office
- NSS Report No. 576, Situation Assessment Survey, 70th Round, 2013
- Agriculture Census 2015-16 and 2010-11, Ministry of Agriculture and Farmers Welfare
- NITI Aayog Discussion Paper 1/2022, "Workforce Changes and Employment: Some Findings from PLFS Data Series"
- Global Yield Gap Atlas (India country page)
- Peer-reviewed hydrology and agronomy literature on groundwater depletion in the Indo-Gangetic plains

This report is Part 1 of the Charitarth Foundation Deep Research Series on Agriculture, Women's Economic Participation and Institutional Architecture in Rural India. It corresponds to Parts I through V of the master research brief.